



Toward Robust and Scalable AI Biomarkers for Systemic Disorders

From Multi-Omics Validation to Wearable Time-Series Foundation Models

Eddie Massey III

Invisible Illness Project · Roeh Health

ME/CFS

POTS

Long COVID

Wearable AI

This Is Personal



Single father — driven to find answers



8+ specialists consulted, no definitive diagnosis



Dismissed, misdiagnosed, told to 'manage symptoms'



Engineering background became research leverage



Founded Invisible Illness Project to serve thousands more like me

"The research gap is a personal emergency."



Why This Problem Is Technically & Clinically Important

The intersection of heterogeneous disease and complex ML



Heterogeneous & Fluctuating

ME/CFS, POTS, Long COVID present differently across patients and across time — one-time testing routinely misses them.



Diagnostic Gap

Millions remain undiagnosed despite high symptom burden and severe functional impairment.



Core ML Challenge

Not simply classification — learning robust representations under missingness, cohort shift, modality imbalance, and weak ground truth.

Why Traditional Biomarker Discovery Struggles Here



Classic Approach

Seeks 1–2 key biomarkers in isolation, treating disease as a simple binary label

Multifactorial Etiology

ME/CFS involves interacting immune, metabolic, microbial, autonomic, and behavioral processes varying across patients.

Attractive but Fragile

BioMapAI explicitly argues that single-outcome / single-modality framing is poorly matched to these diseases.

Overfitting Risk

Especially vulnerable to spurious correlations and non-generalizable 'biomarkers' when systems interact.

Duke/Jackson BioMapAI: Why It Is Novel

AI-driven multi-omics modeling of ME/CFS (2025)

249

participants

153 ME/CFS
96 healthy controls

**4-year
longitudinal**

multi-omics
dataset



Multi-modal integration

Gut metagenomics, plasma metabolomics, immune profiling, blood labs, and clinical symptoms in a single framework.



Multi-output learning

Maps high-dim omics to a matrix of multiple symptom outcomes, capturing both disease-level and symptom-specific biology.



Biological connectivity

Uses SHAP explainability to derive connectivity maps across microbiome, metabolome, and immune features.

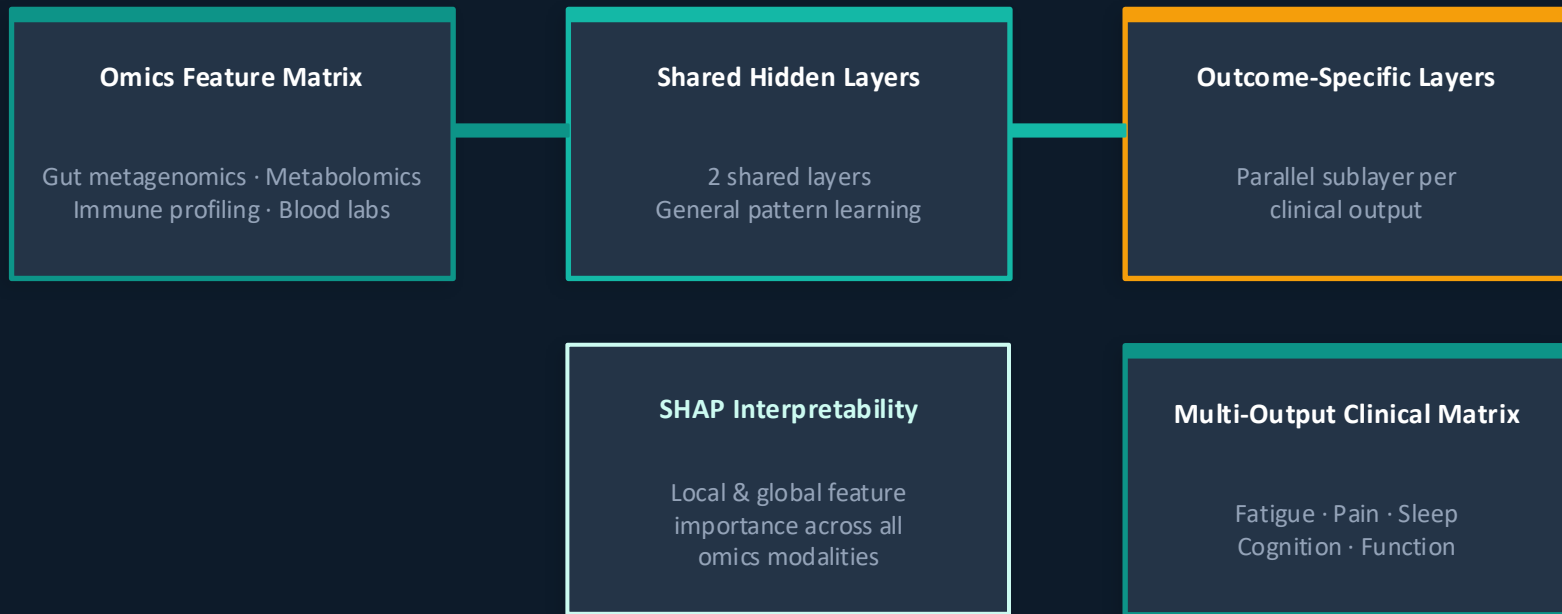


Systems disease framing

One of the strongest attempts to treat ME/CFS as a heterogeneous systems disease, not a simple binary label.

BioMapAI Architecture in Technical Terms

Supervised deep neural network with multi-output symptom-aware design



Key novelty: multi-output symptom-aware multi-omics integration — not just a neural net applied to biology.

BioMapAI Results: What Is Promising

Not just a classifier — a hypothesis-generating multimodal systems model

~0.91

AUC (cross-validation)

disease discrimination

5 of 6

external cohort
comparisons won

Competitive Performance

Better than several baselines across the full omics set and individual modalities in cross-validation.

External Validation

Outperformed comparators in 5 of 6 external cohort comparisons, though external AUCs were notably lower than internal CV.

Biologically Interpretable Links

Identified altered relationships among short-chain fatty acids, BCAAs, mucosal/inflammatory immune cells, and symptom burden.

Hypothesis Generation

BioMapAI is not just a classifier — it provides a systems-level map connecting microbiome, metabolome, and immune biology.

But BioMapAI Is Not the End of the Story

A breakthrough that makes robustness analysis the required next step



Modest dataset size:

249 participants is deep but small by deep learning standards — generalization remains uncertain.



External performance drop:

Cohort differences (technical and clinical) and incomplete feature overlap reduce external AUCs significantly.



Unvalidated symptom biomarkers:

Some symptom-specific outputs were powerful for hypothesis generation but not validated on held-out data.



Supervised label dependency:

Fundamentally tied to available labels and cohort-specific measurement structure — limits prospective deployment.



Feature selection dependency:

Integrated representation depends on prior modality-specific feature selection choices — a hidden sensitivity.

Why a Deeper Robustness & Ablation Study Is the Logical Next Paper

If a multimodal model performs well — what is it actually depending on?

Technical Motivations

- Measure modality dependence — microbiome vs. immune vs. metabolomics?
- Test redundancy — are expensive modalities adding real signal?
- Assess clinical feasibility — can lower-burden inputs proxy high-cost assays?
- Probe overfitting — if accuracy collapses when a modality is perturbed, the biomarker claim is much weaker.

Expected Outputs

- Modality importance ranking
- Redundancy map across assay types
- Robustness curves under degraded input
- Lower-burden proxy candidates
- Feasibility guidance for deployable pipelines

What the Ablation Study Should Do Technically

Proposed experimental design

01 Leave-one-modality-out

Remove microbiome, metabolome, immune, lab, wearable, or symptom-history groups one at a time.

02 Progressive degradation

Simulate lower-resolution assays, noisier measurements, missingness, and reduced feature overlap.

03 Cross-cohort robustness

Evaluate whether modality importance is stable across internal folds, held-out subjects, and external cohorts.

04 Subgroup dependence

Test whether different patient subtypes rely on different modality families.

05 Calibration & uncertainty

Measure whether the model knows when it is out of distribution.

Why This Matters Beyond BioMapAI

The bridge from deep phenotyping to scalable wearable monitoring



Multi-Omics

- Deep biological signal
- Expensive & sparse
- Not continuously available
- Powerful for discovery



Wearable Physiology

- Lower burden & high frequency
- Longitudinal & ecologically valid
- Noisy but continuous
- Essential for flare / recovery dynamics

The most accessible future measurement layer may not be multi-omics — it may be wearable longitudinal physiology.

What Is a Time-Series Foundation Model?

Pretrained on unlabeled temporal data to learn transferable physiological representations

Candidate Pretraining Signals

ECG / RR intervals

PPG

HR / HRV

Accelerometry

EDA

Temperature

Respiration proxies

Sleep & circadian rhythms

Traditional vs. TSFM Approach

Traditional

Hand-engineer HRV / actigraphy features → fit a task-specific model for each outcome

TSFM

Learn broad physiology representation via self-supervised pretraining → fine-tune or probe for specific tasks

Why TSFMs Are Attractive for Systemic Disorders

Directly targeting the bottlenecks in this space



Label Scarcity

Clinical labels are expensive and noisy. Self-supervised pretraining can dramatically reduce labeled-data requirements.



Device Heterogeneity

Pretrained representations may transfer better across sensors and cohorts than hand-engineered pipelines.



Longitudinal Structure

These models exploit rhythms, transitions, missingness patterns, and context at multiple time scales.



Personalization

Embeddings may support patient-specific baselines and drift detection instead of one-size-fits-all thresholds.

Current TSFM Work We Can Build On

Key prior art across wearable biosignal foundation models

Apple

Biosignal Foundation Models:

Self-supervised pretraining on PPG & ECG from ~141K participants shows consumer wearables encode useful health information.

Apple

Behavioral Wearable FMs:

2.5B hours, 162K individuals, strong across 57 health tasks — behavioral representations complement raw-sensor embeddings.

Apple

Accelerometer FM via Distillation:

Lower-fidelity sensors inherit informative structure from richer biosignals — exciting for low-cost digital biomarkers.

Open

PaPaGei PPG Foundation Model:

57,000+ hours of unlabeled PPG from public datasets, strong performance across 20 downstream tasks.

Field

Pulse-PPG:

Field-trained on raw, noisy wearable PPG — emphasizes generalization across field, lab, and clinical environments.

What Is Still Missing for ME/CFS / POTS / Long COVID

The gap is the opportunity

Disease-Specific Dynamics

Most TSFM work is not designed around orthostatic intolerance, PEM-like crashes, or autonomic instability.

Wrong Training Objectives

Models are trained on general wellness or standard clinical benchmarks — not the disease dynamics that matter here.

Field Generalization

Cross-device and field generalization remain hard when signals are noisy and posture/exertion labels are weak or absent.

Needed Representation Learning Emphases

Posture transitions

Exertional context

Circadian state

Symptom-coupled drift

Missingness robustness

Proposed TSFM Research Direction

Pretrain autonomic wearable representations → test on systemic-disorder tasks

Pretraining Signals

- ECG / RR intervals
- PPG
- HR / HRV
- Accelerometry
- EDA
- Temperature
- Respiration proxies
- Sleep / circadian features

Self-Supervised Objectives

- Masked reconstruction
- Contrastive learning across windows
- Motif-aware relative contrastive
- Cross-modal alignment
- Circadian-context prediction
- Posture-transition prediction

Downstream Tasks

- Phenotype discrimination
- Flare prediction
- Recovery trajectory modeling
- Drift detection
- Personalized thresholding

Why TSFMs and BioMapAI Belong in the Same Program

Together, they form a coherent research pipeline



Project A: BioMapAI Robustness

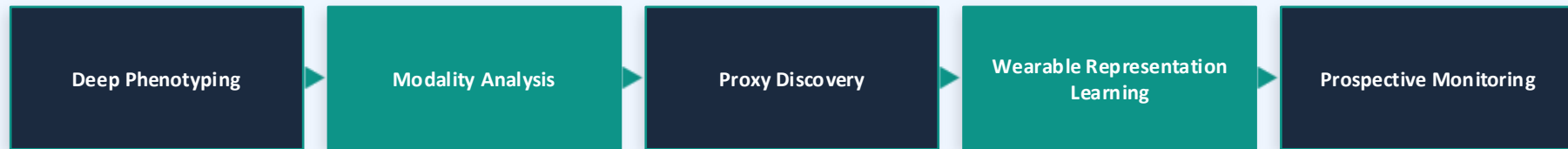
- Identifies which modalities are truly informative
- Tests whether lower-burden substitutes are plausible
- Establishes methodological standards for AI biomarker validation



Project B: Wearable TSFMs

- Provides the candidate substitute measurement layer
- Captures longitudinal physiology unavailable in static multi-omics
- Enables prospective monitoring at scale

Together they support a pipeline:



Strong Candidate Research Questions for PhD Students

Technically rich, scientifically open, clinically meaningful

Q1 Can BioMapAI-like models retain performance under systematic modality dropout?

Q2 Which modality families add unique vs. redundant information?

Q3 Can wearable embeddings recover signal lost when specialty assays are absent?

Q4 Can posture-aware or circadian-aware pretraining improve systemic-disorder phenotyping?

Q5 Can self-supervised embeddings predict flare risk with less labeled data than HRV feature engineering?

Q6 How should uncertainty be estimated when the model is outside the training distribution?

Q7 Can cross-modal distillation help low-cost wearables approximate richer physiological states?

What Makes This a Strong PhD Research Program

Technically real · Scientifically open · Clinically meaningful · Tractable



Technically Real

Multimodal fusion, representation learning, domain shift, uncertainty, interpretability, weak supervision, and causal confounding are all in play.



Scientifically Open

The core digital biomarkers for these diseases are not settled. Every finding is a genuine contribution.



Clinically Meaningful

Underserved populations with real diagnostic gaps. Research here directly translates to patient impact.



Tractable Portfolio

Designed as a portfolio of subprojects rather than one moonshot — permits steady PhD progress with cumulative impact.

Invitation to Collaborate

Invisible Illness Project · Research Opportunities



Multimodal ablation & robustness
pipelines



Wearable time-series foundation models



Physiological signal processing



Domain adaptation across devices



Symptom-aligned labeling & dataset
design



Interpretable flare prediction